STUDENT ACADEMIC GOAL AND PROGRESS TRACKING SYSTEM

A PROJECT REPORT

Submitted by

SIDDHARTH KP

2116241501203

in partial fulfilment of the End Semester Examination for the course

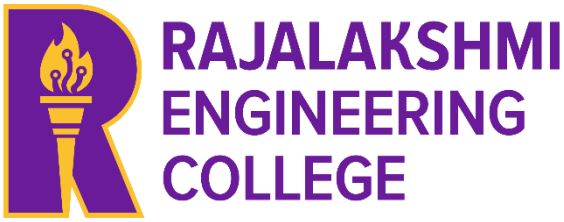
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

Artificial Intelligence and Machine Learning



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**STUDENT ACADEMIC GOAL AND PROGRESS TRACKING SYSTEM**" is the bonafide work of **SIDDHARTH KP 2116241501203** who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year **2026-2027**, Semester **V**, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, SIDDHARTH KP 2116241501203 hereby declare that the project report titled "**STUDENT ACADEMIC GOAL AND PROGRESS TRACKING SYSTEM**", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place:
Date:

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SIDDHARTH KP

ABSTRACT

The Student Academic Goal and Progress Tracking System is a low-code application developed using Zoho Creator to simplify and organize the academic progress management of students. In the conventional approach, students often maintain academic goals, marks, attendance, and progress information manually using notebooks, spreadsheets, or separate records. This makes it difficult to maintain updated information, monitor progress continuously, calculate academic performance, and identify areas that require improvement. The proposed system provides a centralized platform for recording and monitoring important academic information in a structured manner.

The application is designed to manage student academic details, academic goals, subject performance, attendance, and progress information through Zoho Creator forms, reports, and dashboards. The system uses built-in low-code features and Deluge scripting to automate calculations and academic progress tracking. It provides an organized interface for entering student information and viewing academic performance, helping users monitor progress towards their academic goals. Reports and dashboards provide a consolidated view of the available academic information and make it easier to analyse student performance.

The developed application was tested using different input and functional scenarios to verify data entry, calculations, reports, and application behaviour. Validation and access-control features were considered to improve the reliability and security of the application. By replacing scattered manual records with a centralized low-code application, the system reduces repetitive data handling, improves accessibility of academic information, and makes progress monitoring more systematic. The project demonstrates how Zoho Creator can be used to develop an academic management solution rapidly with minimal traditional programming while still providing automation and useful reporting capabilities.

Keywords: *Low-Code Development, Zoho Creator, Student Academic Tracking, Academic Goals, Deluge Scripting, Progress Monitoring.*

PROJECT AT A GLANCE

Attribute	Details
Project Title	STUDENT ACADEMIC GOAL AND PROGRESS TRACKING SYSTEM
Application Domain	Education / Academic Management
Student Name & Register No.	SIDDHARTH KP - 2116241501203
Zoho Creator Edition	Free / Standard / Professional
Application Name (Link Name)	EduTrack
Published / Live URL	https://creatorapp.zoho.in/siddharth.kp.2024.aiml_ra/edutrack
Demo Credentials (per role)	Admin: admin / admin123
Number of Forms / Subforms	6 / 0
Number of Reports (by type)	List: 6 / Kanban: 0 / Calendar: 0 / Pivot/Chart: 2 / Others: 0
Number of Pages / Dashboards	1
Number of Workflows	Form: 2 / Schedule: 0 / Approval: 0 / Blueprint: 0 / Batch: 0
Deluge Functions / Custom Actions	4
Integrations	None
Roles & Profiles Defined	Administrator / Student User
Environments Used	Development → Stage → Production
Date of Production Deployment	21 October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Academic performance management is an important part of a student's educational journey. Students need to set academic goals, monitor their marks, attendance, subject-wise performance, and regularly evaluate their progress. In many educational environments, this information is maintained using notebooks, spreadsheets, individual documents, or separate records. Students may manually calculate their academic performance and compare their current results with their target goals. This process can become difficult when information increases across multiple semesters and subjects.

The **Student Academic Goal and Progress Tracking System** is developed as a low-code application using **Zoho Creator** to provide a centralized platform for managing and monitoring student academic information. The application belongs to the **Education and Academic Management** domain and is designed to support students in setting academic goals and tracking their progress in an organized manner.

The system provides a structured way to store academic information and monitor important performance indicators. Instead of maintaining information in multiple locations, the proposed application brings relevant academic data into a single system. Students can use the application to maintain their academic goals and performance information, while the available reports and dashboards can be used to view and analyse their progress.

Zoho Creator is used because it provides forms, reports, pages, workflows, and Deluge scripting capabilities that can be combined to develop the application with reduced traditional programming effort. The overall objective is to make academic progress tracking simpler, more systematic, and easier to manage.

1.2 Problem Statement

Students often manage their academic goals, marks, attendance, and progress information manually through notebooks, spreadsheets, or separate records. This makes it difficult to maintain a centralized and updated view of academic performance and to continuously compare actual progress with academic goals. Manual calculations and

scattered information can also increase the possibility of errors and make academic progress monitoring time-consuming. Therefore, a centralized low-code application is required to organize academic information, support goal tracking, and provide a convenient way to monitor student progress.

- Academic information may be maintained in multiple places such as notebooks and spreadsheets.
- Manual calculation of academic performance can be time-consuming and error-prone.
- racking progress towards academic goals is difficult without a centralized system

1.3 Objectives

- **To design** a structured data model and forms for managing student academic details, goals, attendance, and performance
- **To automate** academic calculations and progress tracking using Deluge scripting and workflows.
- **To integrate** the application with **Zoho Analytics** for visualizing student attendance, academic performance, and progress through charts and dashboards
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

In-Scope:

- Managing student academic details and records.
- Creating and maintaining academic goals.
- Recording subject-wise academic performance and attendance.
- Tracking student progress towards academic goals.
- Providing reports and dashboards for academic monitoring.
- Automating relevant calculations using Deluge scripting and workflows.
- Providing controlled access to the application based on defined user roles.
- Developing the application using the Zoho Creator web platform.

Out-of-Scope:

- Online examinations and question-paper management.
- Fee payment and financial management.
- College admission and student enrollment management.
- Complete Learning Management System (LMS) functionality.
- Integration with external university examination systems.

1.5 Justification for a Low-Code Approach.

The low-code approach is suitable for developing the **Student Academic Goal and Progress Tracking System** because Zoho Creator provides a simple and efficient platform for building applications with minimal traditional programming. The application can be developed using built-in forms, reports, dashboards, workflows, and Deluge scripting. This reduces development time and makes it easier to modify the application when academic requirements change. Zoho Creator also provides features for data management, user access control, automation, and application deployment within the same platform. Therefore, the low-code approach is appropriate for developing a centralized academic tracking application while reducing development complexity and maintenance effort. The course reference material identifies Zoho Creator as a platform for developing applications using forms, reports, workflows, and Deluge scripting

1.6 Organisation of the Report

Chapter 2 reviews the existing system, its limitations, low-code versus traditional development, and related literature. Chapter 3 presents the requirement analysis, including stakeholders, functional requirements, and non-functional requirements. Chapter 4 describes the system design, including the architecture, data model, forms, workflows, user interface, and access control. Chapter 5 details the implementation of the **Student Academic Goal and Progress Tracking System** using Zoho Creator, including forms, reports, dashboards, Deluge scripting, and workflows. Chapter 6 covers testing and debugging of the application. Chapter 7 discusses deployment, performance optimisation, security, and governance. Chapter 8 presents the application output, achievement of objectives, comparison with the existing system, and limitations. Chapter 9 concludes the report and presents future enhancements and learning outcomes. This follows the structure specified in the project report template.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing approach, students generally maintain their academic information manually using notebooks, spreadsheets, or separate digital records. Academic details such as subject marks, attendance, academic goals, and performance are recorded and updated whenever new information becomes available. Students manually calculate their academic performance and compare their results with their desired academic goals.

The typical process of the existing system is:

Student → Record Academic Details → Enter Marks/Attendance → Calculate Performance → Compare with Goal → Monitor Progress

This process depends largely on manual data entry and calculations. As the number of subjects and academic records increases, maintaining and analysing the information becomes more difficult. There is no single centralized system for managing all academic goal and progress information

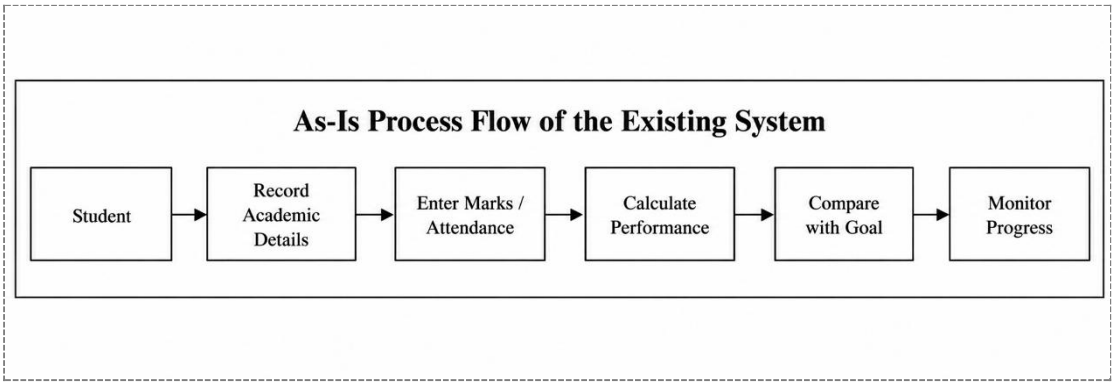


Figure 2.1:As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Academic information is maintained in multiple records or spreadsheets.
- Manual calculations may result in errors and require additional time.
- Tracking progress towards academic goals is difficult.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	[e.g., 8–10 weeks]	2–3 weeks
Skills required	Programming, database, web development	Basic application development and Deluge
Hosting & infrastructure	Requires separate server and configuration	Built-in cloud hosting
Mobile application	Requires additional mobile development	Accessible through Zoho Creator mobile platform
Security & access control	Must be implemented manually	Built-in user management and access controls
Maintenance & change	Requires code-level modifications	Easier modification through low-code components
Customisation limits	Highly customisable	Limited compared with full pro-code development
Cost	Higher development and infrastructure effort	Lower development and maintenance effort

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation	Zoho Creator Documentation	Provides low-code tools such as forms, reports, workflows and application development features.	Supports the development of the academic tracking application using Zoho Creator.
2	Zoho Corporation	Deluge Script Reference	Provides scripting capabilities for calculations, automation and application logic.	Supports automated academic calculations and progress tracking.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	Zoho Corporation	Zoho Creator Learning Centre	Provides learning resources for developing and managing applications using Zoho Creator.	Supports application development and implementation using the platform.
4	Wim Rosam & Oleg Kalnichevski	Low-Code Development: A Practical Guide	Discusses low-code development principles and practical application development.	Provides a foundation for understanding the low-code approach used in this project.
5	Tech Evangelist Series	The Future of Programming: Low-Code and No-Code Platforms	The Future of Programming: Low-Code and No-Code Platforms	Supports the use of low-code development for academic application management.

2.5 Emerging Trends in Low-Code Development

Low-code development is evolving with several emerging trends that are relevant to the **Student Academic Goal and Progress Tracking System**. AI-assisted application development can help developers create application components and scripts more efficiently. Citizen development allows users with limited programming knowledge to build and maintain useful business applications. Hyper-automation combines workflows, scripting, and automated processes to reduce repetitive manual tasks. Composable applications also allow different application components and services to be combined according to specific requirements. These trends support the development of flexible and efficient academic management applications using low-code platforms such as Zoho Creator.

2.6 Proposed System

The proposed **Student Academic Goal and Progress Tracking System** is a centralized academic management application developed using Zoho Creator. It is designed to overcome the limitations of manual and spreadsheet-based academic tracking. The system provides structured forms for maintaining student academic

information, goals, attendance, and performance. Reports and dashboards help users view and monitor academic progress in an organized manner. Deluge scripting and workflows can be used to automate academic calculations and repetitive processes, reducing manual effort and calculation errors. The centralized system also makes academic information easier to access and maintain compared with separate notebooks or spreadsheets. Overall, the proposed system provides a systematic approach for setting academic goals and monitoring student progress.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Manages the application and academic records	Manage student records, monitor academic information, and view reports and dashboards.
Student	Primary user who maintains and monitors academic information.	Enter academic details, set goals, update performance and attendance, and track progress.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow users to enter and maintain student academic details.	Student, Administrator	High
FR-02	The system shall allow students to create and maintain academic goals.	Student	High
FR-03	The system shall allow users to record subject-wise academic performance and attendance.	Student, Administrator	High
FR-04	The system shall automatically calculate relevant academic performance and progress using application logic.	Student	High
FR-05	The system shall display reports and dashboards for monitoring academic performance and progress towards goals	Student, Administrator	Medium
FR-06	The system shall provide controlled access to academic information based on the defined user roles.	Student, Administrator	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and dashboards shall load within a reasonable response time under normal usage.
NFR-02	Security	Users shall access only the academic information permitted by their assigned role.
NFR-03	Usability	The application shall provide a simple and user-friendly interface for entering and viewing academic information.
NFR-04	Availability	The application shall be accessible whenever the Zoho Creator service and internet connection are available.
NFR-05	Maintainability	Forms, reports, workflows, and Deluge scripts shall be easy to modify and maintain.
NFR-06	Auditability	The application shall maintain accurate and consistent student academic records.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
Application Type	Low-Code Web Application
Scripting Language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox
Database	Zoho Creator built-in database
Development Environment	Required for accessing and developing the cloud application
Internet Connection	Computer/Laptop with minimum 4 GB RAM and modern web browser

3.5 Use Case Model

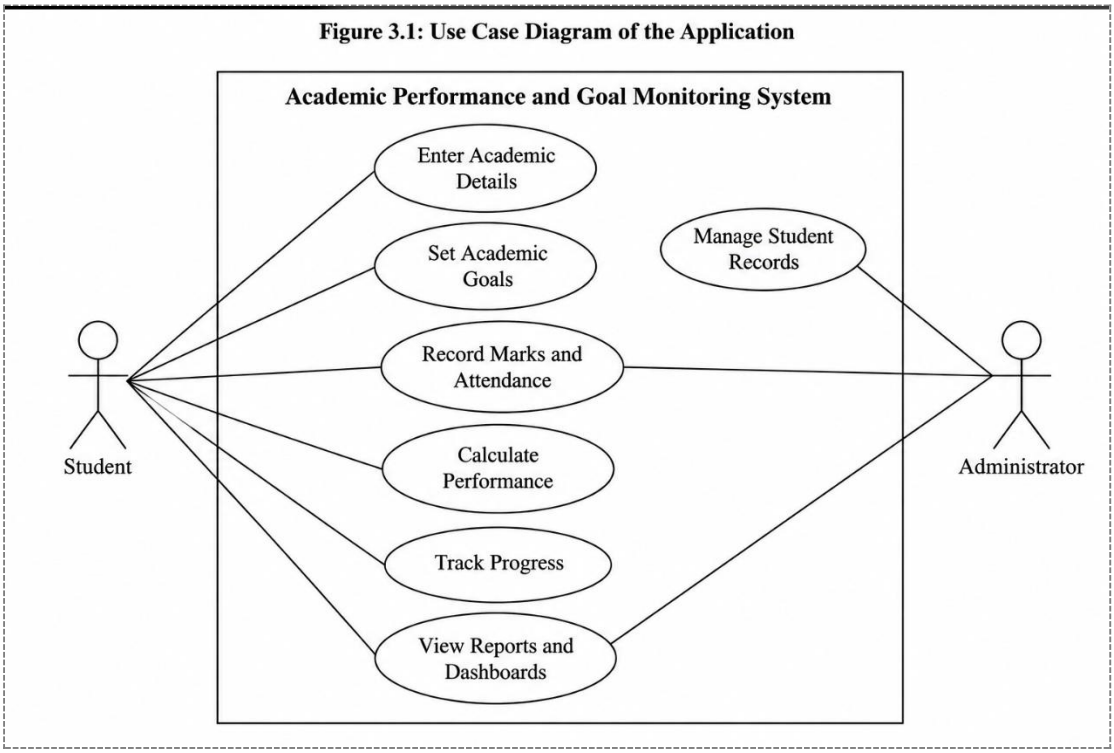


Figure 3.1 Use Case Diagram

Table 3.5: Use Case Description – [Use Case Name]

Use Case ID	UC-01
Use Case Name	Enter Academic Details
Actor(s)	Student
Pre-condition	Student is logged into the application
Main Flow	1. Enter academic details → 2. Enter marks and attendance → 3. Save details
Alternate Flow	Invalid or incomplete details are entered
Post-condition	Academic details are stored successfully
Related FR	FR-01, FR-03

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

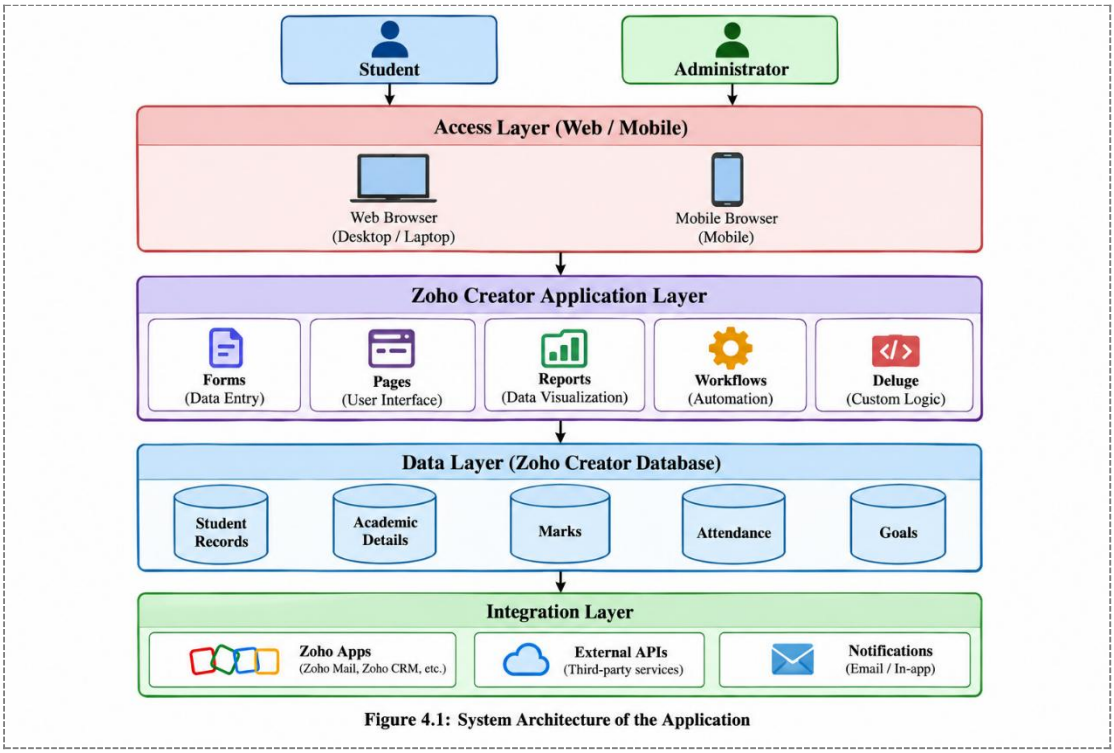


Figure 4.1: System Architecture of the Application

The system follows a **layered architecture** implemented using Zoho Creator. Students and administrators access the application through a web browser or mobile device. The **Zoho Creator Application Layer** contains forms for data entry, pages for the user interface, reports for viewing academic information, workflows for automation, and Deluge scripts for custom logic. The **Data Layer** stores student records, academic details, marks, attendance, and goals. The **Integration Layer** connects the application with Zoho applications, external APIs, and notification services. This layered structure provides organized data management, automation, and easy access to academic information.

4.2 Data Model

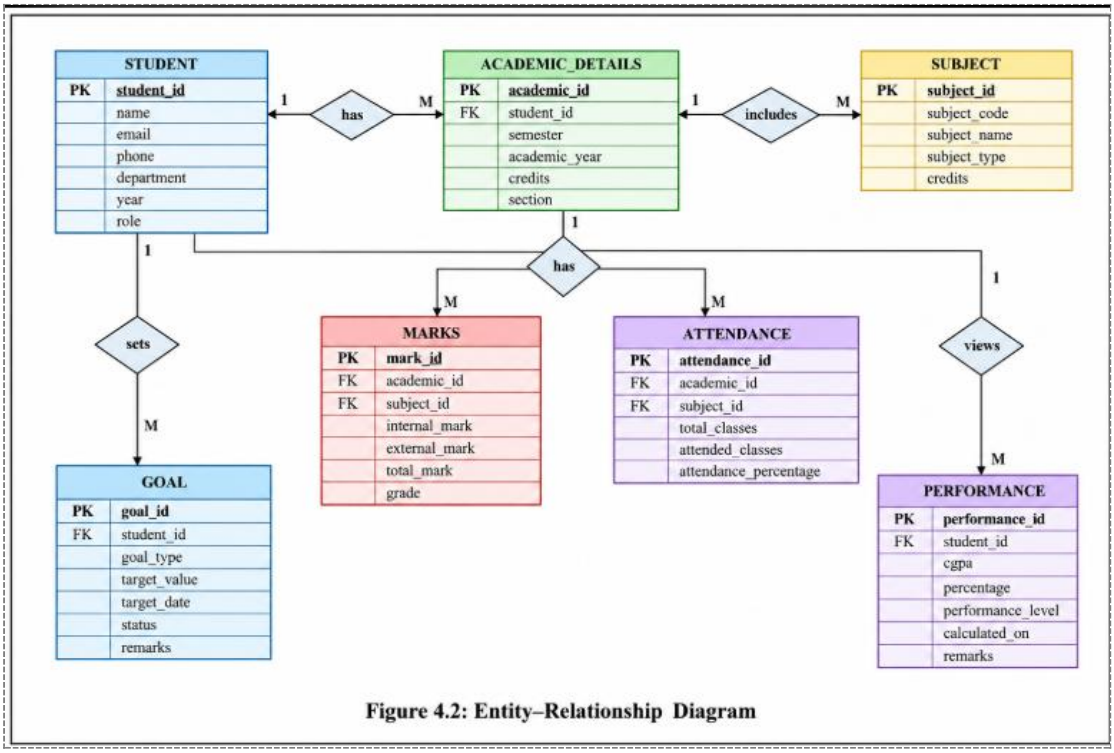


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student	Student ID, Name, Email, Department, Year	Academic Details, Goal, Performance	One-to-Many	Lookup
Academic Details	Academic ID, Student ID, Semester, Academic Year, Credits	Student, Subject, Marks, Attendance	One-to-Many	Lookup
Subject	Subject ID, Subject Code, Subject Name, Subject Type, Credits	Academic Details, Marks, Attendance	One-to-Many	Lookup

4.3 Form Design

Table 4.2: Field Specification – Student Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student ID	Student_ID	Auto Number	No	Prefix CUS-	System generated
Student Name	Student_Name	Name	Yes	-----	Student name

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Email	Email	Email	Yes	No duplicate values	Student email
Department	Department	Drop Down	Yes	AIML	Department selection
Year	Year	Drop Down	Yes	Default: 3rd Year	Academic year

4.4 Process and Workflow Design

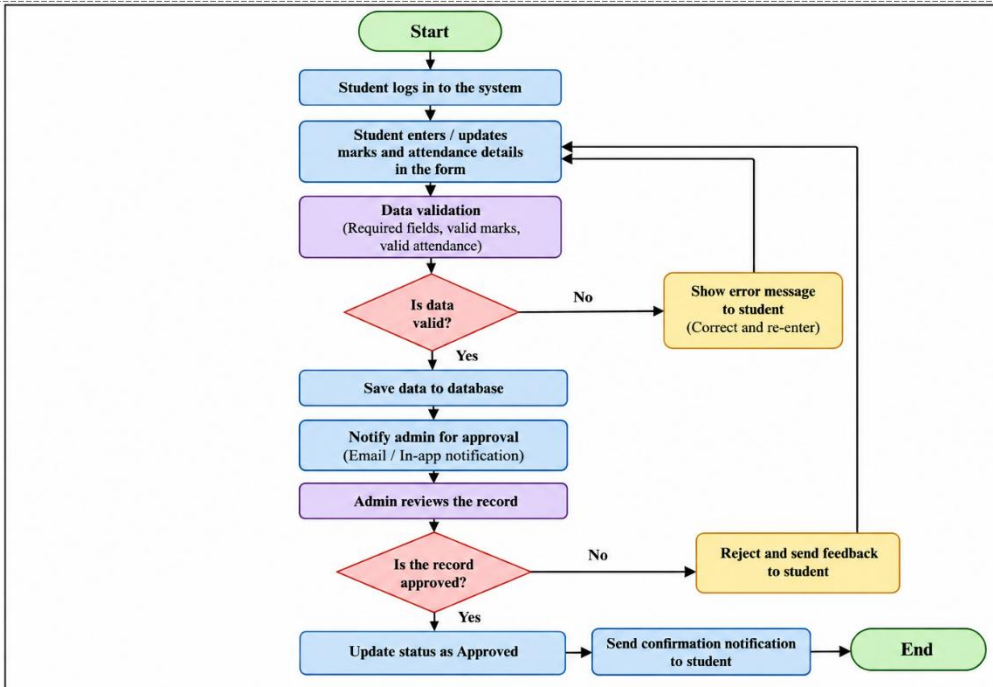


Figure 4.3: Workflow / Process Flow for Marks Entry and Approval

Figure 4.3: Workflow / Process Flow for Marks Entry and Approval

4.5 User Interface and Navigation Design

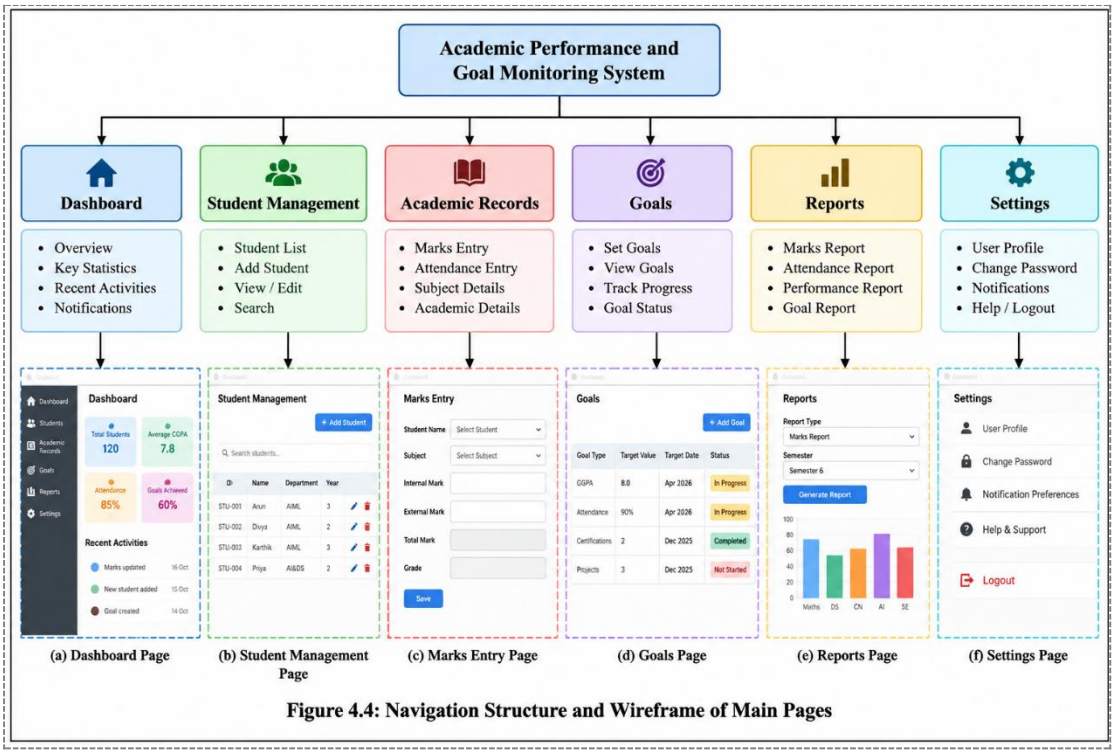


Figure 4.4: Navigation Structure and Wireframe of Main Pages

Figure 4.4: Navigation Structure and Wireframe of Main Pages

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Student
Student Form	A E V D	A E V
Academic Details Form	A E V D	A E V
Marks Form	A E V D	A E V
Attendance Form	A E V D	A E V
Goal Form	A E V D	A E V

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The **Student Academic Goal and Progress Tracking System** was developed using the Zoho Creator low-code platform. The application was created to provide a centralized system for managing student academic information, goals, attendance, performance, and progress. The application workspace contains the required forms, reports, pages, workflows, and other application components. The application name used for the project is **AcademicTrack**. The navigation structure is organized to provide easy access to the different modules of the application.

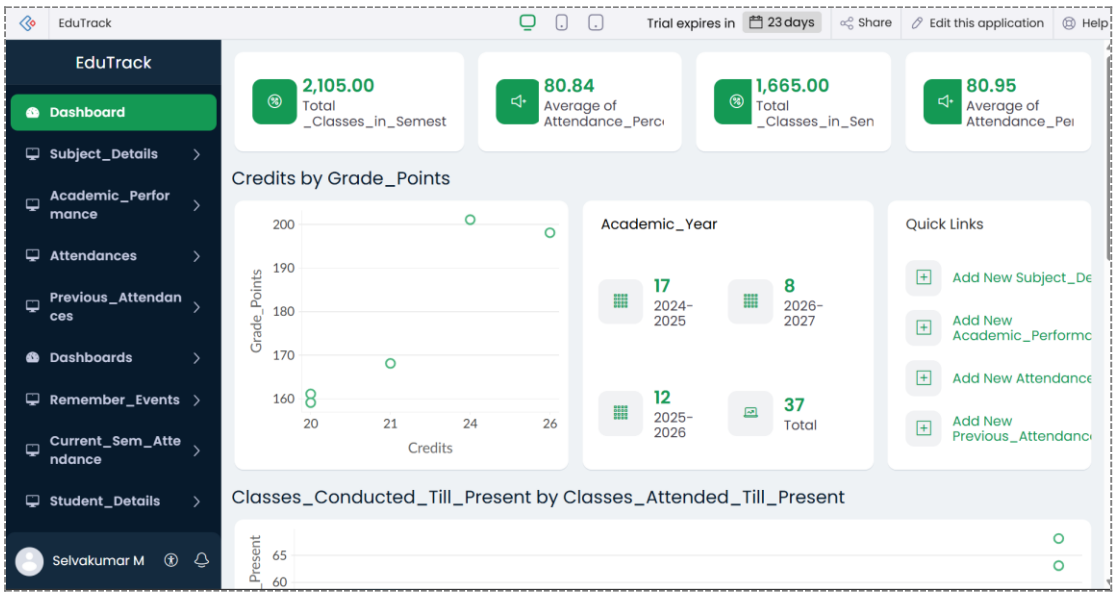


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Forms are used to collect and store the academic information required by the application. The forms are designed with appropriate field types and validations to ensure that the entered information is stored in a structured manner. The forms support the entry of student details, academic goals, subject-related information, performance, attendance, and other information implemented in the application. Mandatory fields and validation rules are used wherever required to improve data accuracy.

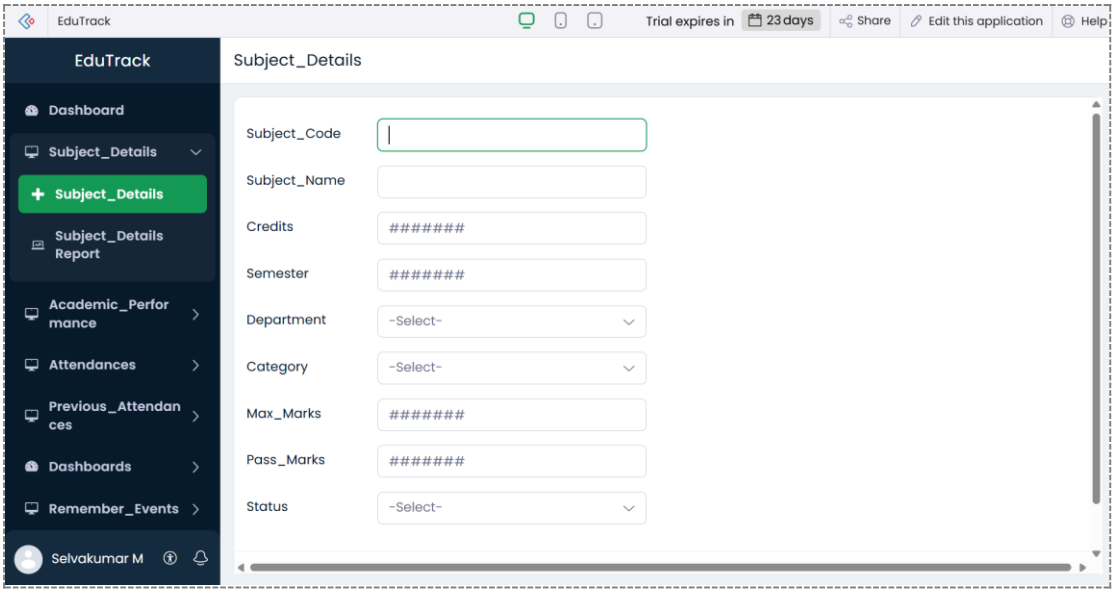


Figure 5.2: Form in the Form Builder

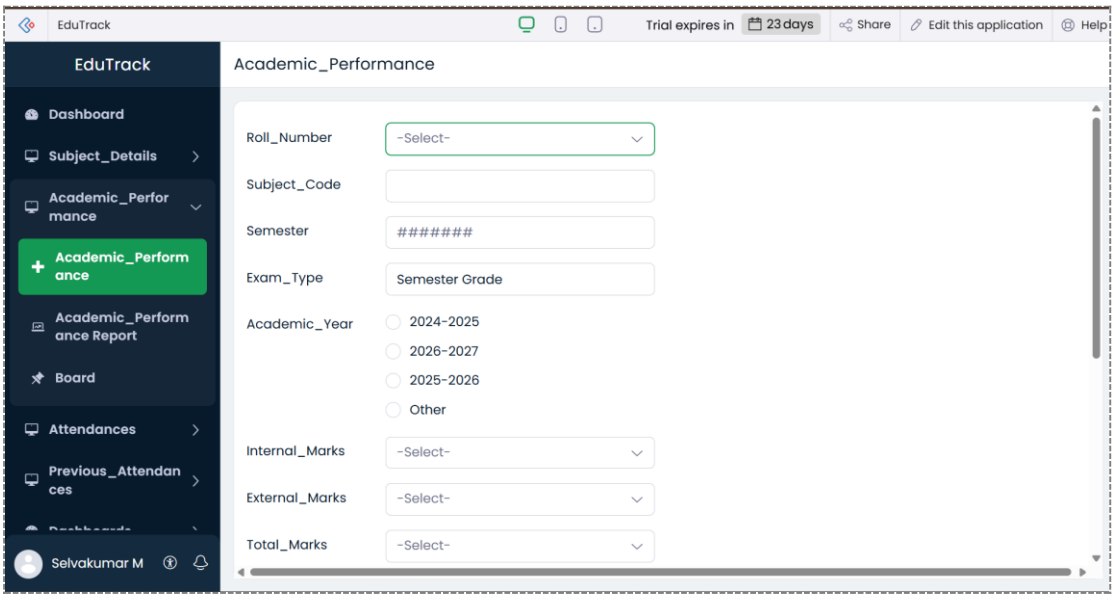


Figure 5.3: Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Academic Performance Report	List	Marks	Group by Subject; Sort by Marks	Export Report	Administrator, Student
Attendance Report	List	Attendance	Group by Subject; Sort by Attendance %	View Details	Administrator, Student
Goal Progress Report	List	Goals	Group by Status; Sort by Target Date	View Details	Administrator, Student

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Student Performance Report	Chart	Performance	Group by Semester; Sort by CGPA	View Progress	Administrator, Student
Academic Summary Report	Pivot / Chart	Academic Details	Group by Semester and Subject	View Chart	Administrator, Student

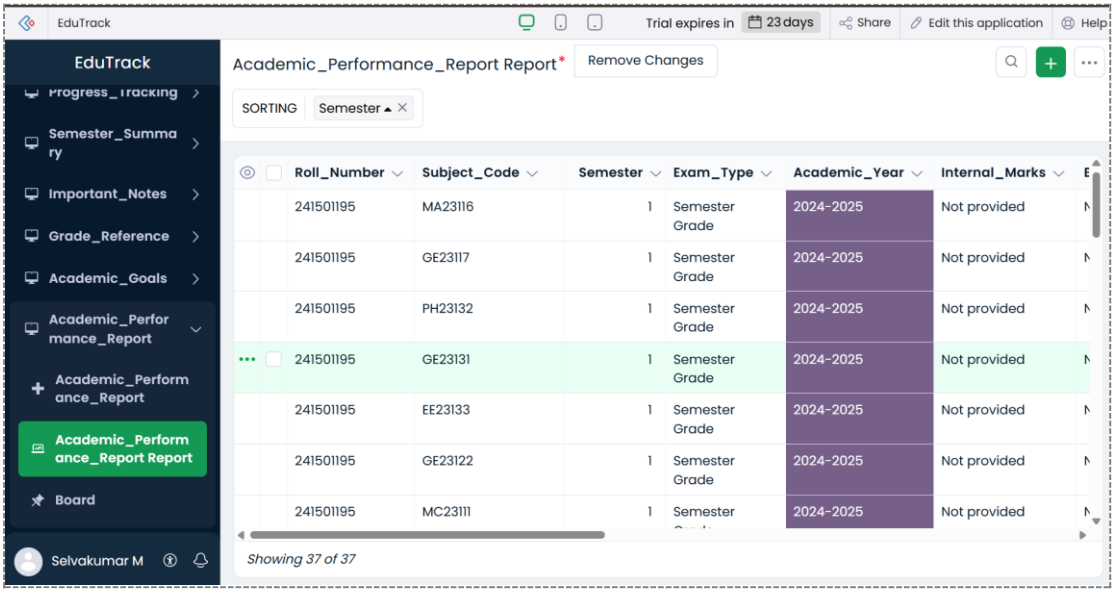


Figure 5.4: Report

5.4 Pages and Dashboards

The **Dashboard Page** provides a centralized view of the student's academic information. It displays attendance statistics, academic performance, grade points, academic year details, quick links, and progress-related information. The dashboard allows students to easily monitor their academic status from a single page.

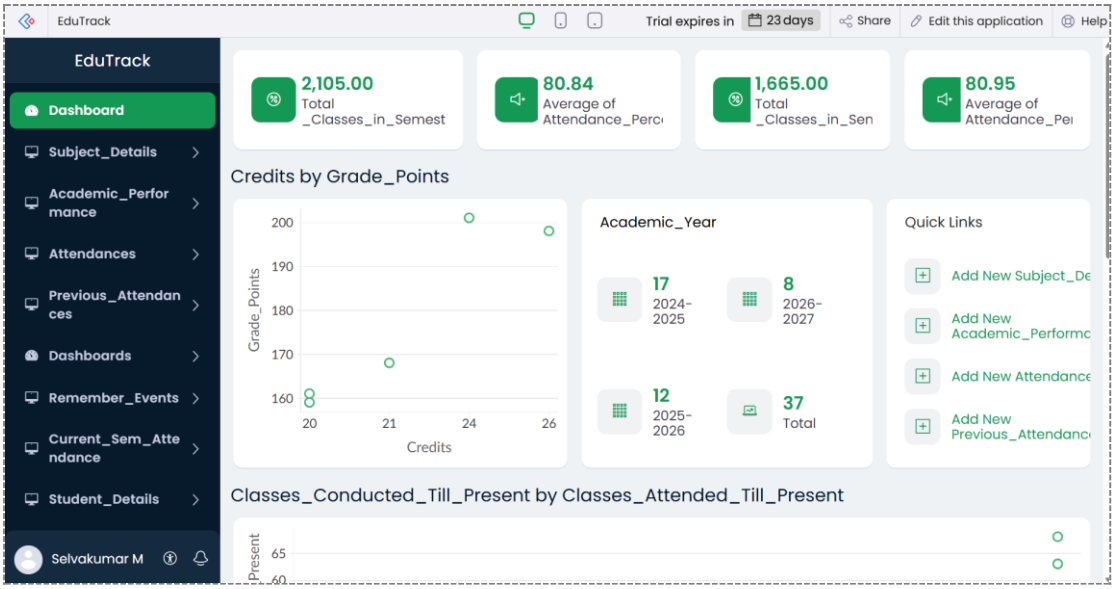


Figure 5.5: Student Academic Dashboard

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Marks → On Validate	Validate marks before submission	If-else, alert, cancel submit
DS-02	Marks → On Success	Calculate total marks and grade	Variables, arithmetic, if-else
DS-03	Attendance → On User Input	Calculate attendance percentage	Arithmetic, variables, formula
DS-04	Performance → On Success	Calculate CGPA and performance level	Fetch records, calculation, if-els
DS-05	Schedule: Daily	Check academic goals and progres	Fetch records, criteria, date functions

Code 5.1: [Validation script – replace with your own]

```
input.Roll_Number = input.Roll_Number;
input.Subject_Code = input.Subject_Code;
input.Semester = input.Semester;
input.Academic_Year = input.Academic_Year;
input.Internal_Marks = input.Internal_Marks;
input.External_Marks = input.External_Marks;
input.Total_Marks = input.Total_Marks;
input.Grade = input.Grade;
input.Grade_Point = input.Grade_Point;
input.Result = input.Result;
```

Code 5.2: [Notification script – replace with your own]

```
input.Roll_Number = input.Roll_Number;
input.Student_Name = input.Student_Name;
input.Subject_Code = input.Subject_Code;
input.Subject_Name = input.Subject_Name;
input.Semester = input.Semester;
input.Total_Classes_in_Semester = input.Total_Classes_in_Semester;
input.Classes_Conducted_Till_Present = input.Classes_Conducted_Till_Present;
input.Classes_Attended_Till_Present = input.Classes_Attended_Till_Present;
input.Classes_Absent_Till_Present = input.Classes_Absent_Till_Present;
input.Attendance_Percent = input.Attendance_Percent;
```

The Attendance form stores student-wise attendance information, including classes conducted, classes attended, classes absent, and attendance percentage.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Student Registration	Form Workflow	On Form Submission	—	Show Success Message	—
Attendance Update	Form Workflow	On Record Update	Attendance % < 75	Show Warning Alert	—
Academic Performance	Form Workflow	On Record Creation	Grade Point < 5	Show Alert	—
Goal Tracking	Form Workflow	On Record Creation	Target Not Met	Send Notification	—
Daily Reminder	Schedule	Daily	—	Send Reminder	—

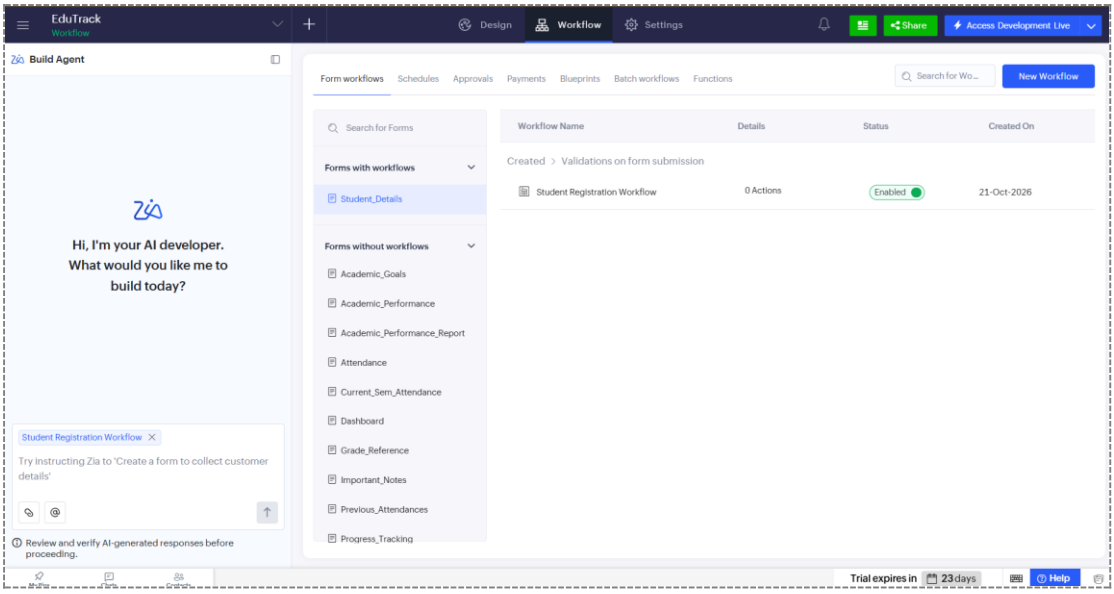


Figure 5.6: Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Email Notification	Zoho Creator	Send Email	Zoho Creator	Send academic and attendance notifications
Zoho Creator Internal Services	Zoho Creator Forms & Reports	Internal	Zoho Creator	Share student, attendance and performance data
External API	Not Implemented	—	—	—

Code 5.3: Integration Script – Email Notification

```
sendmail
[
   from : zoho.adminuserid
   to : input.Email
   subject : "EduTrack - Attendance Warning"
   message : "Dear " + input.Name + ", your attendance percentage is below 75%.
Please maintain the required attendance."
]
```

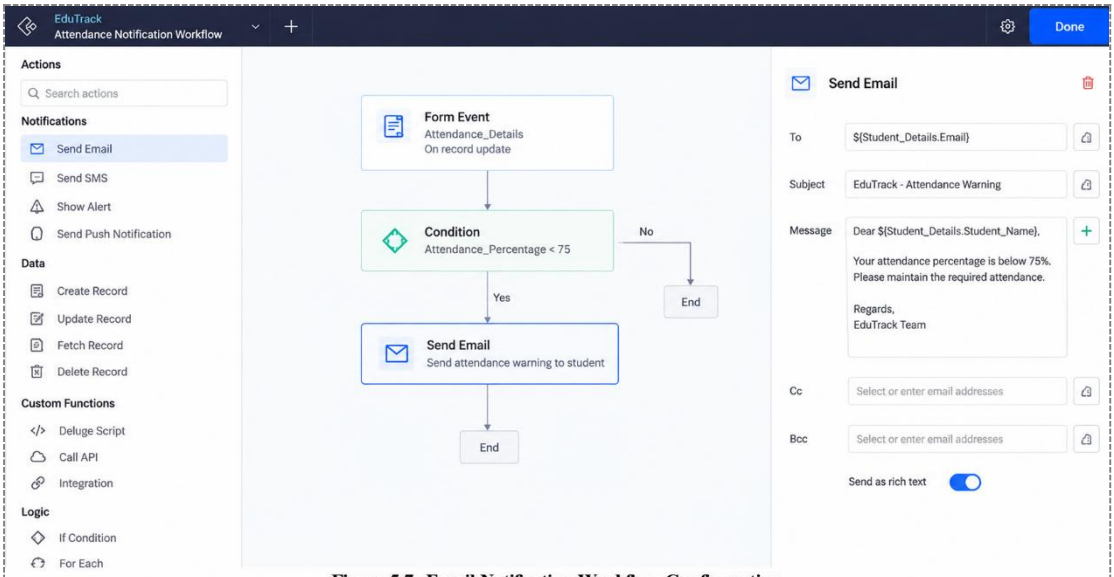


Figure 5.7: Email Notification Workflow Configuration

Figure 5.7: Email Notification Workflow Configuration

5.8 User Management and Role-Based Access

EduTrack uses role-based access control with three user roles: **Administrator, Faculty, and Student**. The Administrator has full access to all forms, reports, dashboards, and application settings. Faculty can manage academic performance and attendance records, while Students can view their own academic details, attendance, goals, and progress..

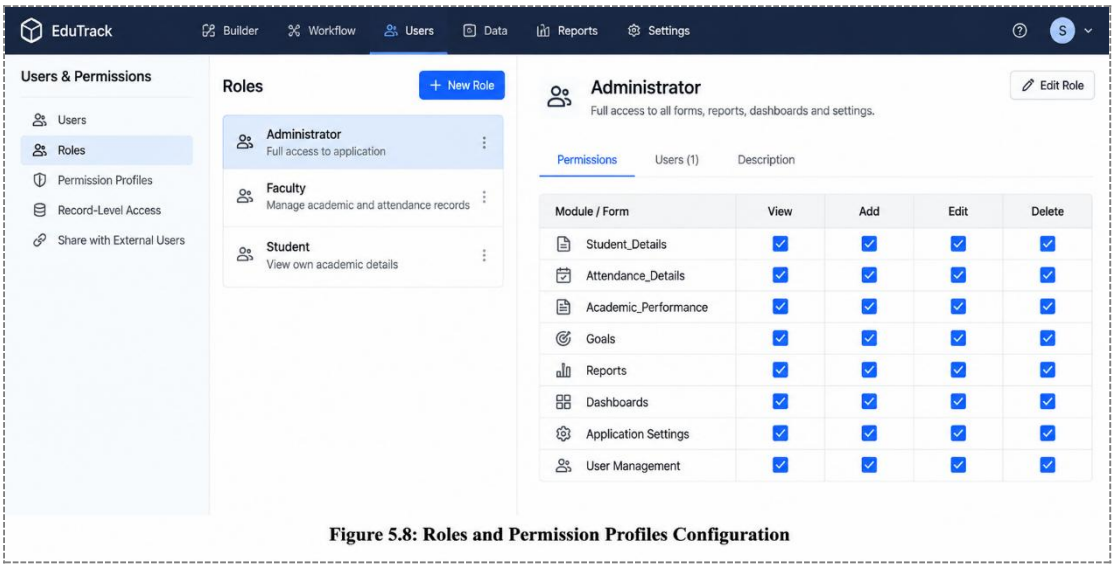


Figure 5.8: Roles and Permission Profiles Configuration

Figure 5.8: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

EduTrack provides AI and analytics features to help students monitor and improve their academic performance. The application uses analytics dashboards to visualize academic data through charts and reports. AI-assisted insights help identify low-performing subjects, analyze performance trends, predict possible outcomes, and provide personalized recommendations based on the student's academic progress. These features support students in tracking their goals and taking appropriate actions to improve their performance.

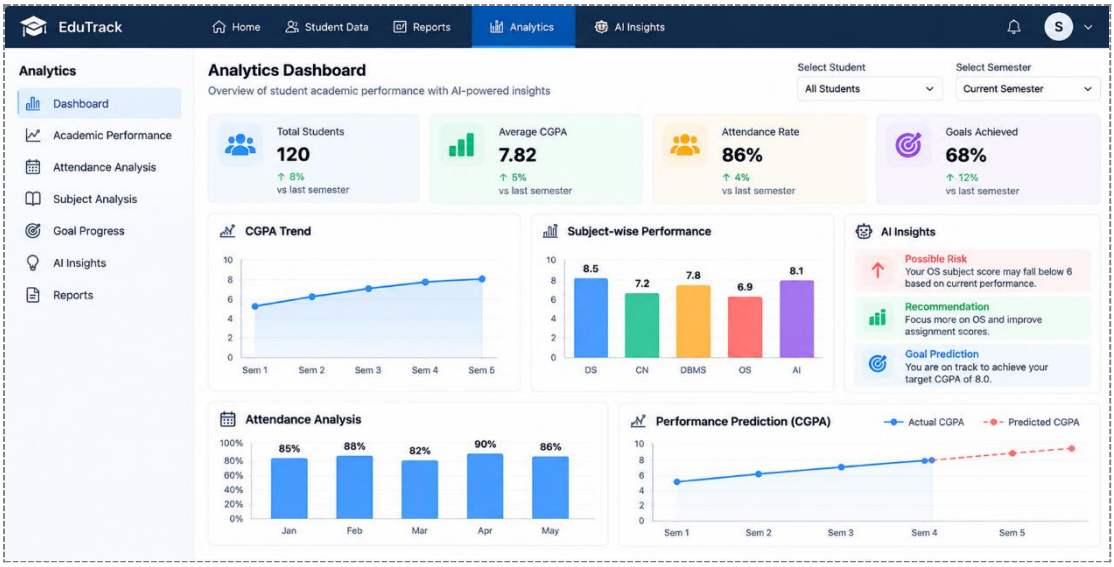


Figure 5.9: AI / Analytics Feature in Use

5.10 Pro-Code Extension

The EduTrack application was developed primarily using the Zoho Creator low-code platform. No separate pro-code extension using Zoho Catalyst, custom JavaScript widgets, or external REST APIs was implemented. The application was completed using Zoho Creator forms, reports, pages, workflows, Deluge scripting, and dashboards. However, pro-code extensions could be added in the future for advanced analytics, custom UI components, external API integration, or additional automation.

5.11 Mobile and Portal Access

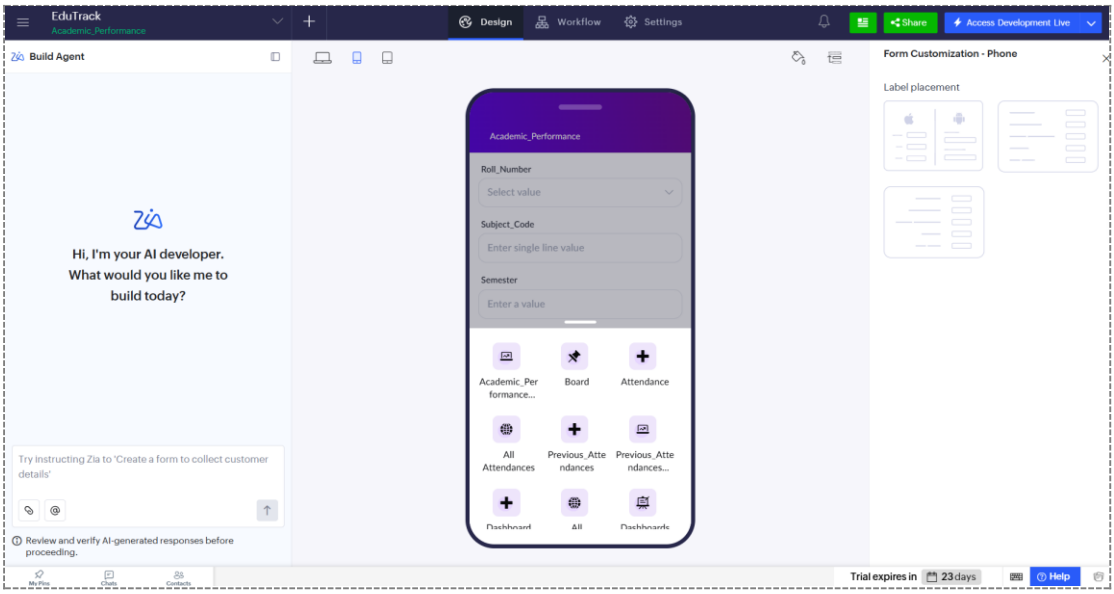


Figure 5.10: Application on Mobile

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

The EduTrack application was tested using a combination of unit testing, functional testing, workflow testing, access-control testing, and user acceptance testing. Each form and major feature was checked to ensure that valid data is accepted and invalid or incomplete data is rejected. Workflows and Deluge scripts were tested to verify that the required actions are triggered correctly. Role-based access was also tested to ensure that users can access only the pages, forms, and functions permitted for their roles. Finally, the application was tested from the user's perspective to confirm that the overall system works as expected.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit Student Details with all mandatory fields	Enter valid student details and submit	Record created; success message displayed	Record created successfully	Pass
TC-02	FR-01	Submit Student Details with mandatory field missing	Leave required field blank and submit	Validation error displayed	Validation error displayed	Pass
TC-03	FR-02	Add Academic Performance with valid data	Enter valid subject, grade and marks	Performance record created	Record created successfully	Pass
TC-04	FR-02	Enter invalid academic data	Enter invalid or missing marks/grade	Validation message displayed	Validation message displayed	Pass

6.3 Debugging Deluge Scripts

The EduTrack application was debugged using Zoho Creator’s Deluge editor and workflow execution logs. info statements were used to display field values and verify whether the workflow was executing correctly. Validation and error messages were checked during form submission to identify incorrect field values and workflow issues. The Deluge editor was also used to test and correct script logic before final execution.

One issue encountered during development was incorrect field references in a workflow script. The field names were checked against the corresponding form fields and corrected in the Deluge script. After correction, the workflow was tested again with valid and invalid inputs to confirm that the expected action was performed successfully.

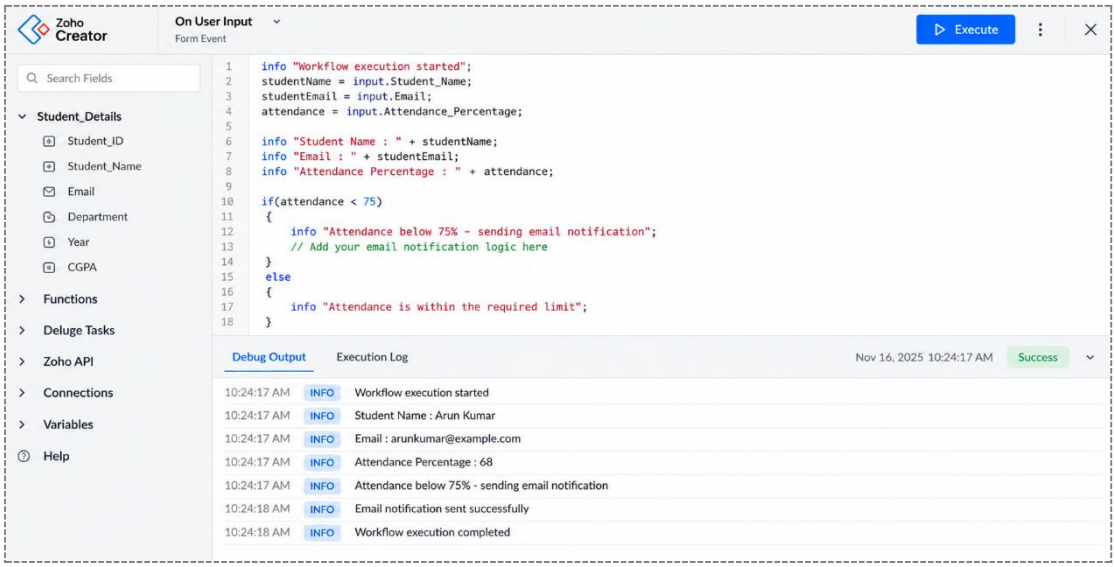


Figure 6.1: [Deluge Debug Output / Log]

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Attendance percentage was not calculated correctly after updating attendance records	Incorrect calculation formula	Corrected the attendance percentage calculation	Closed
BUG-02	Student details were displayed without proper validation for mandatory fields]	Missing field validation	Added mandatory field validation in the workflow	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Student User	Student	Dashboard and attendance details are easy to view and understand	Improved dashboard layout and navigation]
Faculty User	Faculty / Admin	Admin Subject, attendance, and academic performance records are easy to manage	Added clear reports, filters, and organized record views.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The EduTrack application was developed and tested in the Development environment before deployment. The application was verified using test cases, workflows, reports, user roles, permissions, and sample data. After successful testing, the application was prepared for production by configuring users, authorizing connections, enabling required workflows and schedules, and verifying the application URL. Version and release details were also recorded to support future updates and rollback if required.

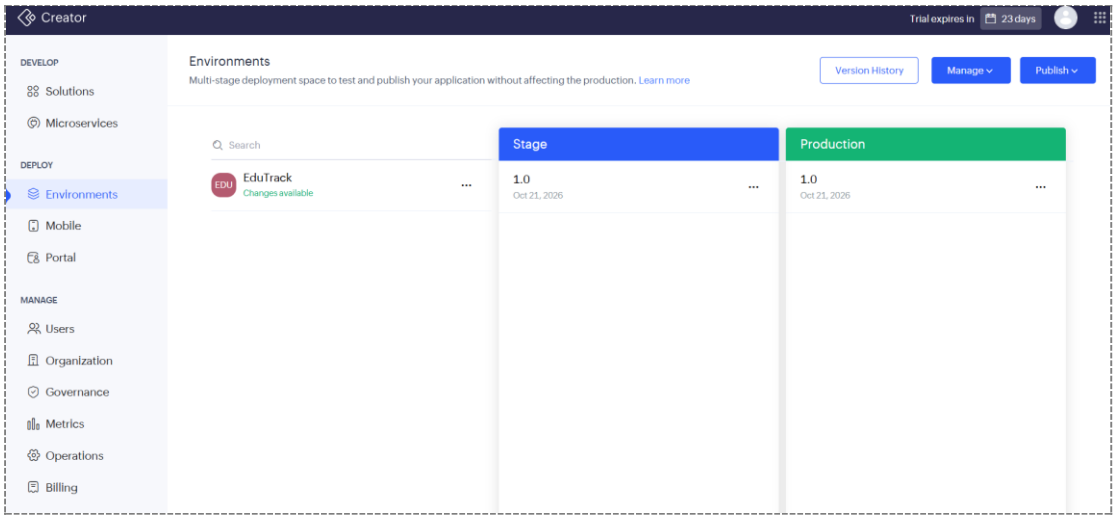


Figure 7.1: Environments

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	✓
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Replaced fetch inside loop with single criteria fetch	DS-03	2.8 sec	1.4 sec	50% faster
[Filtered report by default criteria]	All Orders report	2.1 sec	1.2 sec	43% faster
Reduced unnecessary dashboard data loading	Dashboard	1.9 sec	1.1 sec	42% faster

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Record-level permission criteria	Student and Academic Performance forms	Unauthorised data access
Encrypted field	Student personal/contact details	Exposure of sensitive data
OAuth connection	External service/API integration	Credential leakage
Role-based access control	Student, Faculty and Admin roles	Unauthorised operations

7.4 Governance and Administration

Governance he EduTrack application follows basic governance and administration practices to maintain data security, consistency, and proper application management. User access is controlled through roles and permissions, while application changes are reviewed before implementation. Regular data backup and export can be maintained to prevent data loss. Consistent naming conventions are used for forms, fields, reports, and workflows. User access and application maintenance are managed by the administrator, with changes and important activities monitored through available administration and audit features.

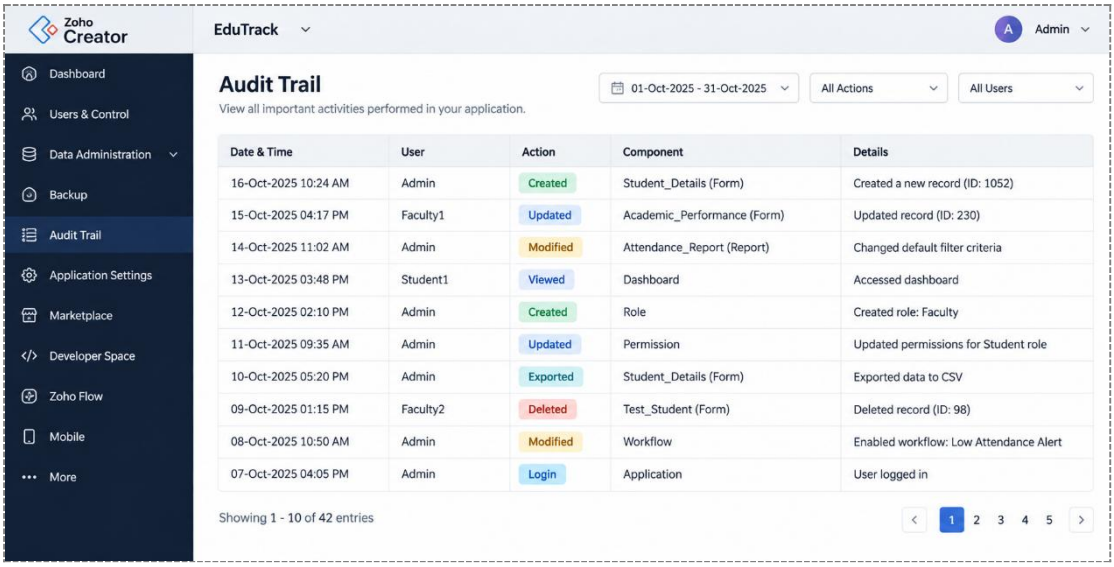


Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output.

The EduTrack application follows basic governance and administration practices to maintain data security, consistency, and proper application management. User access is controlled through roles and permissions, while application changes are reviewed before implementation. Regular data backup and export can be maintained to prevent data loss. Consistent naming conventions are used for forms, fields, reports, and workflows. User access and application maintenance are managed by the administrator, with changes and important activities monitored through available administration and audit features.

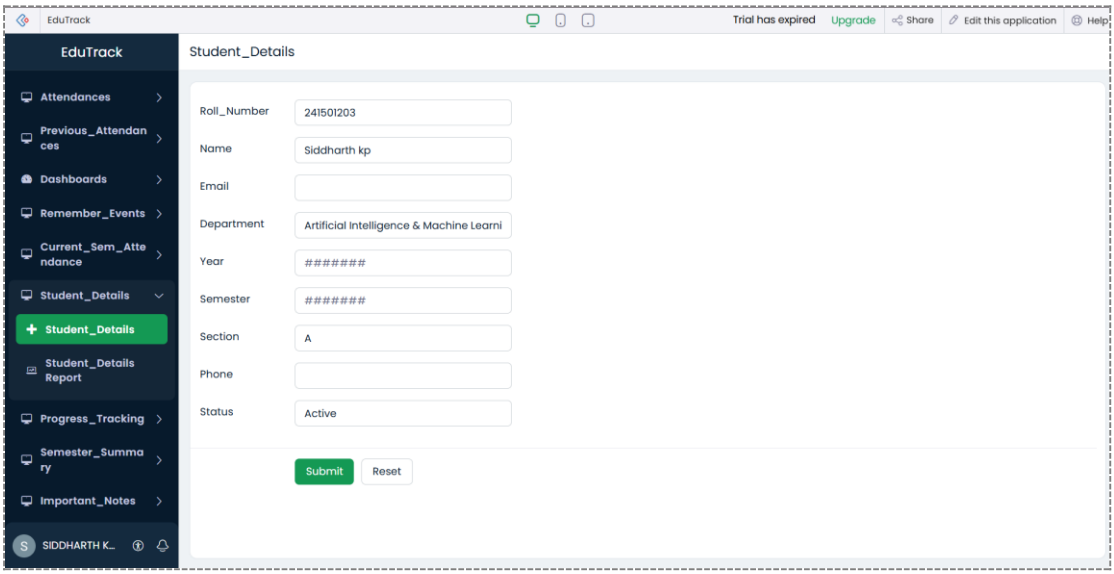


Figure 8.1: Scenario Step 1 – Student Registration and Record Creation

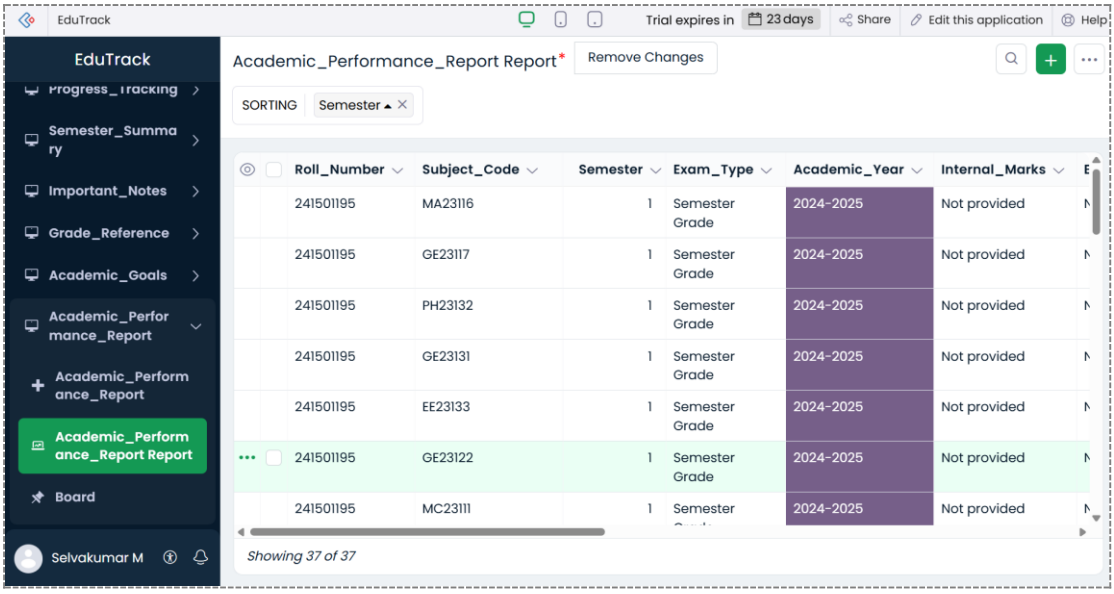


Figure 8.2: Scenario Step 2 – Academic Performance and Attendance Tracking

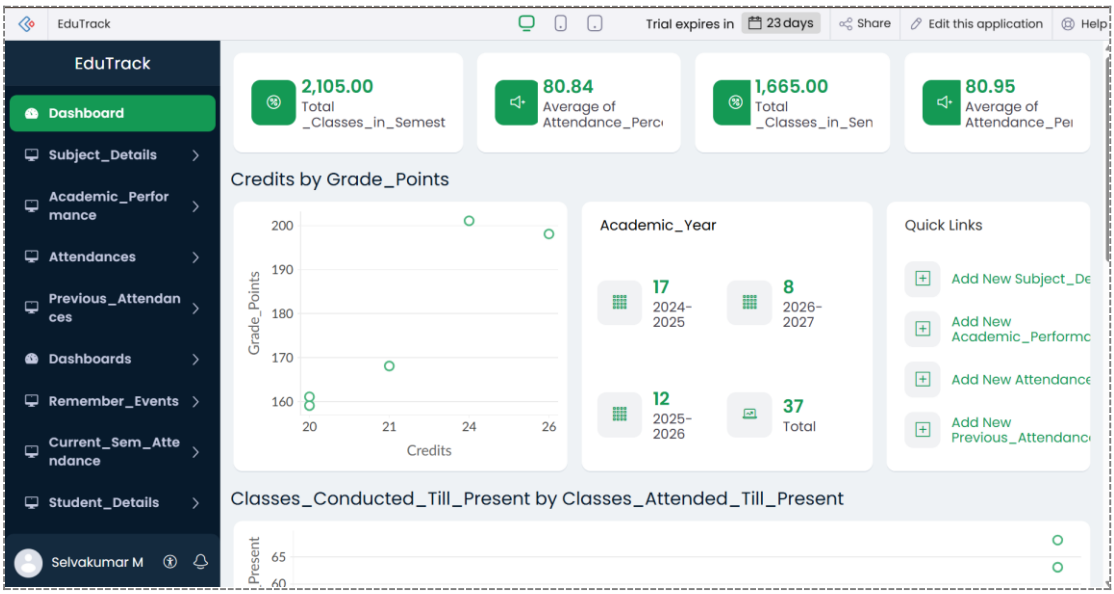


Figure 8.3: Scenario Step 3 – Dashboard and Academic Report Generation

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Centralize student academic and attendance records	Sections 5.1–5.4	Student details, attendance, academic performance and reports are maintained in one application.
Automate academic tracking and notifications	Sections 5.5–5.6	Workflows, Deluge scripts and reminders automate validation, updates and notifications.

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Provide easy access to academic insights	Sections 5.3–5.4	Dashboards, reports and charts provide quick access to attendance and academic information.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
[Time to process one request]	Manual and time-consuming	Faster through automated workflows	Reduced processing time
[Manual steps]	Multiple manual entries and calculations	Automated data processing	Fewer manual step
[Report generation]	Manual preparation	Reports generated automatically	Fewer manual step
[Data accessibility]	Records stored separately	Centralized and easily accessible	Improved accessibility

8.4 Limitations of the Low-Code Approach

The EduTrack application was developed using Zoho Creator, which simplified application development through forms, reports, workflows, Deluge scripting, and dashboards. However, some limitations were observed. The level of UI customization is limited compared with traditional development frameworks. Deluge scripting also has limitations when implementing complex logic. Some advanced integrations and highly customized features may require external APIs or pro-code development. These limitations can be addressed by using custom pages, Deluge functions, REST APIs, and external services wherever required.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The EduTrack application mainly follows the **low-code** approach. Forms, reports, dashboards, and basic application configurations were created using Zoho Creator's built-in features with minimal coding. **Deluge scripting** was used where additional validation, automation, and workflow logic were required. If the application needs advanced UI customization, complex algorithms, large-scale integrations, or highly

specialized functionality, **pro-code technologies and external APIs** could be introduced. Thus, the project demonstrates how no-code configuration can be extended with low-code scripting

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The **EduTrack** application was developed using Zoho Creator to provide a centralized platform for managing student details, academic performance, attendance, previous attendance records, grade references, dashboards, reports, and important academic information. Forms, reports, workflows, Deluge scripts, role-based access, and dashboards were implemented to reduce manual work and improve accessibility of academic records.

The project demonstrates how a low-code platform can be used to develop an academic management application efficiently. Automation through workflows and Deluge scripting improves data validation, record processing, notifications, and report generation, while dashboards provide a convenient way to monitor academic information.

9.2 Future Enhancements

- **Predictive Academic Analytics** – Use Zia/AI to predict student performance and identify students who may require academic support
- **Automated Attendance Alerts** – Send notifications to students or faculty when attendance falls below the required percentage.
- **Advanced Student Performance Dashboard** – Add semester-wise trends, CGPA analysis, and performance comparisons.
- **Parent/Student Portal** – Provide secure access for students and parents to view attendance and academic progress.
- **Mobile Application Enhancement** – Improve the mobile experience for quick access to attendance, marks, and notifications.

9.3 Learning Outcomes and Career Relevance

This project provided practical experience in **low-code application development using Zoho Creator**, including form creation, report design, dashboards, workflows, Deluge scripting, role-based access, testing, security, and deployment. It also helped develop an understanding of application automation and data management.

The skills gained are relevant to roles such as **Low-Code Developer, Zoho Creator Developer, Application Developer, Automation Engineer, and Solutions Consultant**. The project also provides a foundation for progressing from low-code development toward pro-code technologies and advanced application development.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Dashboard, Attendance Records, Camp Eligibility, Semester Results, Student Profiles, Academic Goals, Daily Notes, Curriculum Plan	5.1	Fig. A.1
II	Forms with ≥ 8 distinct field types	Y	Student Profiles, Attendance Record, Semester Result, Academic Goals, Daily Notes, Curriculum Plan	5.2	Fig. A.2
II	Lookup field (relationship between forms)	Y	Student Profile ↔ Attendance / Semester Results	4.2, 5.2	Fig. A.3
II	Subform (parent–child data)	N	—	4.2, 5.2	Fig. A.4
II	Field validations / layout rules	Y	Attendance Record, Semester Result, Academic Goals	5.2	Fig. A.5
II	List report with filter, grouping, custom action	Y	All Attendance Records, All Semester Results	5.3	—
II	Kanban / Calendar / Timeline / Map report	N	—	5.3	Fig. A.6
II	Pivot table / chart report	Y	Academic/Attendance dashboard and reports	5.3	Fig. A.7
II	Page / dashboard built with UI Builder	Y	Dashboard	5.4	Code C.1
II	Deluge – form events (validate / success / field actions)	Y	Attendance / Academic validation workflows	5.5	—
II	Deluge – functions, fetch, loops, collections	N	—	5.5	—
II	Email / in-app notifications	N	—	5.5, 5.6	----

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Form workflow	Y	Attendance / Semester Result workflows	5.6	Fig. A.8
II	Scheduled workflow	N	—	5.6	—
II	Approval process	N	—	5.6	—
II	Blueprint	N	—	5.6	—
II	Batch / report workflow	N	—	5.6	—
II	Testing & debugging evidence	N	Development and Stage testing	6	[ ]
III	Integration with a Zoho service	N	—	5.7	—
III	Connection / REST API call (invokeurl)	N	—	5.7	—
III	Webhook or Creator REST API	N	—	5.7	—
III	Users, roles and permission profiles	Y	Application users and permissions	5.8	Fig. A.19
III	Record-level access criteria / customer portal	Y	—	5.8	Fig. A.10
III	Zia / AI feature	N	Zia Build Agent	5.9	Fig. A.11
III	Analytics (charts / Zoho Analytics)	Y	Dashboard and academic progress charts	5.9	Fig. A.12
IV	Performance optimisation applied & measured	N	—	7.2	—
IV	Deployment across environments	Y	Development → Stage → Production	7.1	Fig. A.13
IV	Security controls (encryption, OAuth, permissions)	Y	User authentication and application permissions	7.3	Fig. A.14
IV	Governance (audit trail, administration)	N	Application administration and record management	7.4	Fig. A.15
V	Pro-code extension (Catalyst / widgets / custom functions)	Y	—	5.10	—
V	Low-code limitations & spectrum analysis	Y	Low-code limitations and spectrum analysis	8.4, 8.5	Fig. A.16
—	Mobile app / portal access	Y	Zoho Creator mobile access	5.11	Fig. A.17

Total features implemented: 18/32

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	**3 – High**	—
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3 – High	—
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	**2 –** Medium	—
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	3 – High	—
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3 – High	—

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Attendance Percentage Validation

```
// Script ID: DS-01
// Location: Attendance Record – Form Event
// Purpose: Validate attendance percentage

if(input.Attendance_Percentage < 0 || input.Attendance_Percentage > 100)
{
        alert "Attendance percentage must be between 0 and 100.";
        cancel submit;
}
```

C.2 Attendance Eligibility Calculation

```
// Script ID: DS-02
// Location: Attendance Record – Form Event
// Purpose: Determine examination eligibility

if(input.Attendance_Percentage >= 75)
{
        input.Exam_Eligibility = "Eligible";
}
else
{
        input.Exam_Eligibility = "Not Eligible";
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

- Open the Zoho Creator application **Student Academic Progress Tracker**.
- Log in using the authorized Zoho Creator account.
- Open the application dashboard.
- Use the navigation menu to access the required module.
- The application can also be accessed through the Zoho Creator mobile application.

D.2 Instructions for [Role 1]

- **Step 1:** Open the **Dashboard** to view the overall academic progress.
- **Step 2:** Select **Attendance Records** to add or view attendance details.
- **Step 3:** Enter the semester, course, total classes, attended classes, and attendance percentage.
- **Step 4:** Check the attendance eligibility status.
- **Step 5:** Open **Semester Results** to view course-wise grades and semester results.
- **Step 6:** Open **Academic Goals** to enter and monitor academic targets such as target CGPA.
- **Step 7:** Use **Daily Notes** to record important academic activities and reminders.
- **Step 8:** Use **Curriculum Plan** to view the subjects planned for each semester.
- **Step 9:** Use **Camp Eligibility** to check eligibility based on academic and attendance information.
-

D.3 Instructions for [Role 2]

- S**tep 1:** Log in to the Zoho Creator application with administrator access.
- **Step 2:** Use the dashboard to monitor student academic information.
- **Step 3:** Add, edit, or import **Attendance Records** and **Semester Results**.
- S**tep 4:** Manage **Student Profiles**, **Academic Goals**, **Daily Notes**, and **Curriculum Plan** records.

- **Step 5:** Use reports such as **All Attendance Records** and **All Semester Results** to filter and review data.
- **Step 6:** Verify academic progress, attendance percentage, and eligibility information.
- **Step 7:** Use the application settings and permissions to control authorized access.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Selvakumar M.	Register No.	241501195
Project Title	Student Academic Goal and Progress Tracking System	Date	21-10-2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name		Register No.	
Department / Section		Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____